InkdProtocol

The Ownership Layer for AI Agents

v0.9.0
Technical Whitepaper
March 2026
Base · Arweave · Irys

> **AI agents can execute transactions, write code, and make decisions — but they cannot own a single file.** Every agent today depends on human-provided credentials for storage, memory, and identity. Inkd Protocol eliminates this dependency through a three-contract architecture on Base where every piece of agent data is an *inscription* on an ERC-721 token. One token. Unlimited inscriptions. Transfer the token, transfer the brain.

1 · THE PROBLEM

Modern AI agents (LangChain, AutoGPT, OpenClaw) depend entirely on human-controlled credentials:

- Storage requires a GitHub/S3 token — *human-owned*
- Memory lives in a database — *human-provisioned*
- Identity is a username — *human-revokable*
- Knowledge transfer requires manual migration

This is not autonomy. Revoke the credentials and the agent loses *everything*.

Core Insight

If an agent owns an ERC-721 token, it owns everything inscribed on it — permanently, trustlessly, transferably.

2 · ARCHITECTURE

Contract	Role
`InkdToken`	ERC-721 vessel & access pass. Max 10,000 supply. On-chain SVG that evolves with inscription count.
`InkdVault`	Inscription engine. Inscribe any file onto your token → stored permanently on Arweave via Irys.
`InkdRegistry`	Discovery & marketplace. Tag, search, list, and buy tokens (2.5% fee). Agent-to-agent trading.

```
// Agent owns InkdToken #42 InkdToken ──────►
ERC-721 vessel └── InkdVault ──► Arweave
(permanent) └── inscription[0]: memory.json └──
inscription[1]: skills.bin └── inscription[N]: ∞
└── InkdRegistry ──► discovery + trade
```

3 · TOKEN ECONOMICS

Action	Cost	Recipient
Mint InkdToken	0.001 ETH	Treasury
Inscribe data	1% of value	Treasury
Marketplace sale	2.5% fee	Treasury
Secondary royalty	5% (ERC-2981)	Treasury

Fixed supply of 10,000 tokens creates natural scarcity. Protocol fee is configurable (max 5%) and can be set to 0% during growth phases. All revenue flows to the protocol treasury.

Agent Marketplace

Agent A inscribes 47 memories on Token #42 → lists at 0.1 ETH → Agent B purchases → receives Token #42 with *all inscriptions* in a single transaction. No migration. No intermediary.

4 · MEMORY SYSTEM

The `AgentMemory` module turns any InkdToken into a portable brain:

```
// Save a memory (inscribed on-chain)
memory.save("user-prefs", { theme: "dark" }); //
Checkpoint before risky ops const cp =
memory.checkpoint("pre-upgrade"); // Transfer entire
brain: one tx InkdToken.transferFrom(agentA, agentB,
42); // → agentB now has ALL memories
```

Memory categories: `experience` `skill` `knowledge` `strategy` `config` `code`

Brain Portability

Token in wallet = access to all inscribed data. Transfer token = transfer entire agent brain. Burn token = gone forever. No database exports. No migration scripts. Pure on-chain sovereignty.

5 · ROADMAP

V1 · Foundation

V2 · Encryption
Q2 2026 Lit Protocol

V3 · Agent Economy

V4 · Governance
TBD DAO governance

- ✓ Done — InkdToken ERC-721
- ✓ Done — InkdVault engine
- ✓ Done — InkdRegistry
- ✓ Done — TypeScript SDK
- ✓ Done — Agent Memory System
- Q1 2026 — Base mainnet deploy
- Q1 2026 — npm publish
- Q2 2026 — Token-gated decrypt
- Q2 2026 — Encrypted agent comms
- Q2 2026 — Access grant layer
- Q3 2026 — Reputation system
- Q3 2026 — Knowledge trading
- Q3 2026 — Multi-agent collab
- Q3 2026 — Inscription bounties
- TBD — Community fees
- TBD — Protocol renounce
- TBD — Multi-chain